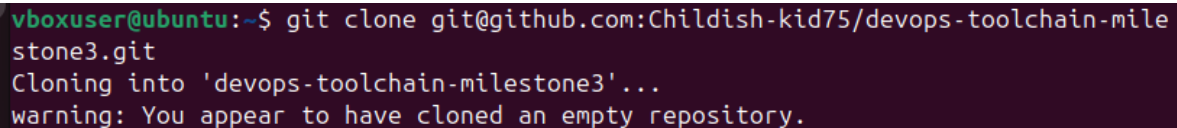


Repository Setup

Objective: Clone the empty GitHub repository to the local machine

Command:

```
git clone git@github.com:Childish-kid75/devops-toolchain-milestone3.git
```



```
vboxuser@ubuntu:~$ git clone git@github.com:Childish-kid75/devops-toolchain-milestone3.git
Cloning into 'devops-toolchain-milestone3'...
warning: You appear to have cloned an empty repository.
```

Figure 1: Successful repository clone (empty repository warning is expected)

Create Project Structure

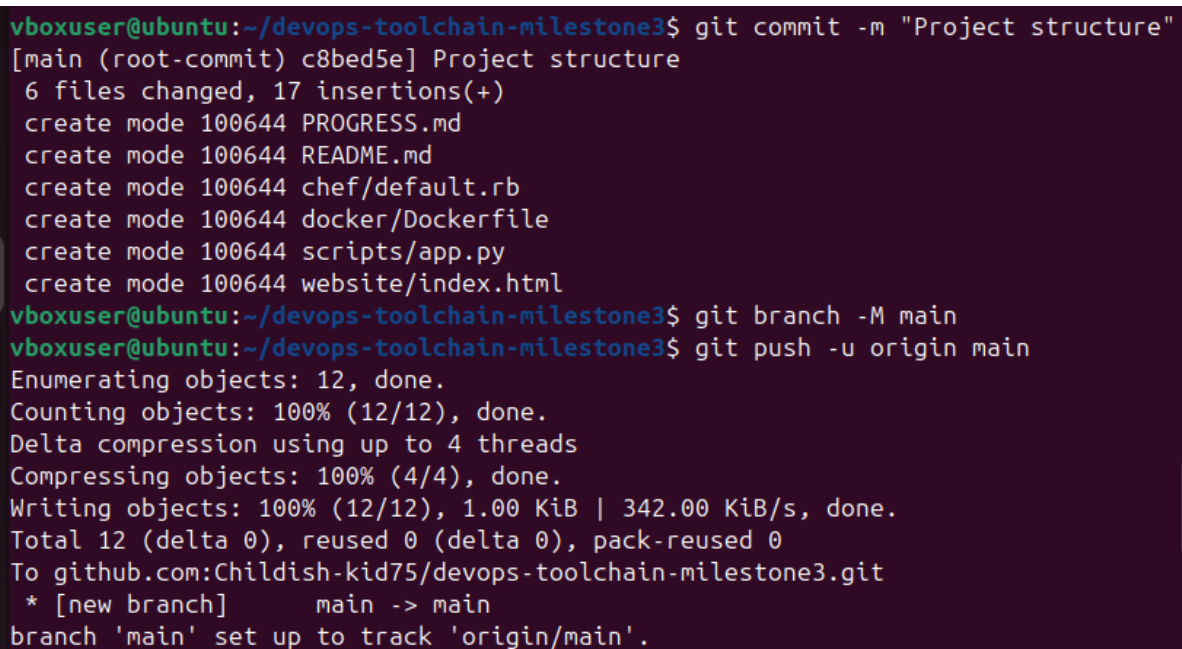
Objective: Create initial project folders and files

Commands:

```
mkdir -p scripts docker chef website docs
```

```
echo "# DevOps Toolchain Project" > README.md
```

Initial commit:



```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git commit -m "Project structure"
[main (root-commit) c8bed5e] Project structure
 6 files changed, 17 insertions(+)
 create mode 100644 PROGRESS.md
 create mode 100644 README.md
 create mode 100644 chef/default.rb
 create mode 100644 docker/Dockerfile
 create mode 100644 scripts/app.py
 create mode 100644 website/index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git branch -M main
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (12/12), 1.00 KiB | 342.00 KiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:Childish-kid75/devops-toolchain-milestone3.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Figure 2: Initial project structure committed to main branch

Create Individual Folders

Objective: Create folders for all team members' Part B exercises

Command:

```
mkdir -p  
individuals/{tshiamo,darius,ditheto,sizakele,nthailane,temosho,christiaan,reghardt,steven}/{docker,git,chef}
```

Output:

```
vboxuser@ubuntu:~$ cd ~/devops-toolchain-milestone3  
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git checkout main  
Switched to branch 'main'  
Your branch is up to date with 'origin/main'.  
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git pull origin main  
From github.com:Childish-kid75/devops-toolchain-milestone3  
* branch            main      -> FETCH_HEAD  
Already up to date.  
vboxuser@ubuntu:~/devops-toolchain-milestone3$ mkdir -p individuals/{tshiamo,darius,ditheto,sizakele,nthailane,temosho,christiaan,reghardt,steven}/{docker,git,chef}
```

Figure 3: Individual folders created for all team members

Create Individual Branches

Objective: Create personal branches for each team member

Commands:

for name in tshiamo darius ditheto sizakele nthailane temosho christiaan reghardt steven; do

git checkout -b \$name-partb

git push origin \$name-partb

git checkout main

done

Output:

```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ for name in tshiamo darius ditheto sizakele nthailane temosho christiaan reghardt steven; do  
> git checkout -b $name-partb  
> git push origin $name-partb  
> git checkout main  
> done  
Switched to a new branch 'tshiamo-partb'  
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0  
remote:  
remote: Create a pull request for 'tshiamo-partb' on GitHub by visiting:  
remote:      https://github.com/Childish-kid75/devops-toolchain-milestone3/pull/new/tshiamo-partb  
remote:  
To github.com:Childish-kid75/devops-toolchain-milestone3.git  
* [new branch]      tshiamo-partb -> tshiamo-partb  
A       individuals/README.md  
Switched to branch 'main'  
Your branch is up to date with 'origin/main'.  
Switched to a new branch 'darius-partb'  
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0  
remote:  
remote: Create a pull request for 'darius-partb' on GitHub by visiting:  
remote:      https://github.com/Childish-kid75/devops-toolchain-milestone3/pull/new/darius-partb  
remote:  
To github.com:Childish-kid75/devops-toolchain-milestone3.git  
* [new branch]      darius-partb -> darius-partb  
A       individuals/README.md
```

Figure 4: Personal branches created for all team members

Add Placeholder Files

Objective: Preserve directory structure in Git

Command:

```
find individuals -type d -exec touch {}/.gitkeep \;
```

```
git add .
```

```
git commit -m "Create individual folders and branches for Part B exercises"
```

Output:

```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ find individuals -type d -exec touch {}/.gitkeep \;
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git add .
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git commit -m "Create individual folders and branches for Part B exercises"
[main 69b4d8e] Create individual folders and branches for Part B exercises
35 files changed, 18 insertions(+)
create mode 100644 individuals/.gitkeep
create mode 100644 individuals/README.md
create mode 100644 individuals/christiaan/.gitkeep
create mode 100644 individuals/christiaan/chef/.gitkeep
create mode 100644 individuals/christiaan/docker/.gitkeep
create mode 100644 individuals/christiaan/git/.gitkeep
create mode 100644 individuals/ditheto/.gitkeep
create mode 100644 individuals/ditheto/chef/.gitkeep
create mode 100644 individuals/ditheto/docker/.gitkeep
create mode 100644 individuals/ditheto/git/.gitkeep
create mode 100644 individuals/nthailane/.gitkeep
create mode 100644 individuals/nthailane/chef/.gitkeep
create mode 100644 individuals/nthailane/docker/.gitkeep
create mode 100644 individuals/nthailane/git/.gitkeep
create mode 100644 individuals/reghardt/.gitkeep
create mode 100644 individuals/reghardt/chef/.gitkeep
create mode 100644 individuals/reghardt/docker/.gitkeep
create mode 100644 individuals/reghardt/git/.gitkeep
create mode 100644 individuals/sizakele/.gitkeep
create mode 100644 individuals/sizakele/chef/.gitkeep
```

```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 984 bytes | 246.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Childish-kid75/devops-toolchain-milestone3.git
c8bed5e..69b4d8e  main -> main
```

Figure 5: Placeholder files added and pushed to main branch

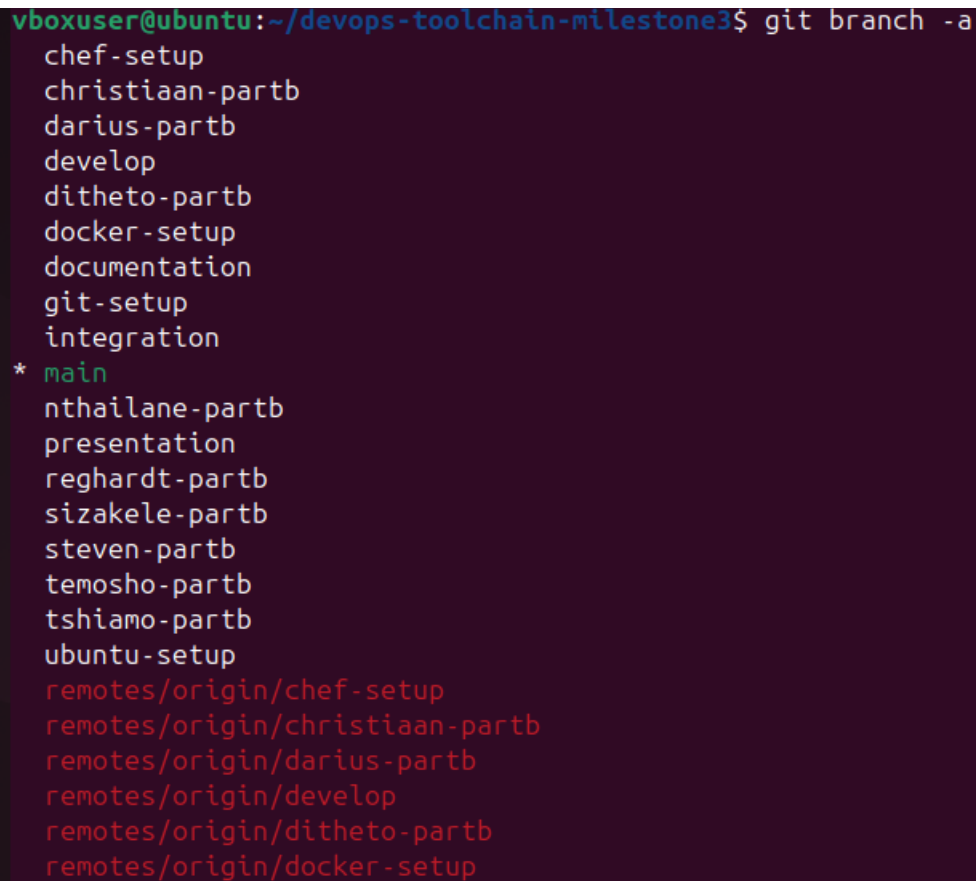
Verify Branches

Objective: Confirm all branches were created successfully

Command:

```
git branch -a
```

Output:



```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git branch -a
  chef-setup
  christiaan-partb
  darius-partb
  develop
  ditheto-partb
  docker-setup
  documentation
  git-setup
  integration
* main
  nthailane-partb
  presentation
  reghardt-partb
  sizakele-partb
  steven-partb
  temosho-partb
  tshiamo-partb
  ubuntu-setup
remotes/origin/chef-setup
remotes/origin/christiaan-partb
remotes/origin/darius-partb
remotes/origin/develop
remotes/origin/ditheto-partb
remotes/origin/docker-setup
```

Figure 6: All team branches visible (chef-setup, docker-setup, and individual -partb branches)*

Docker Exercise

Objective: Create a Python script and Dockerfile

Files Created:

app.py:

```
print("=" * 50)

print("DevOps Practical Assignment")

print("Docker Container Running Successfully!")

print("Executed by: Tshiamo")

print("=" * 50)
```

Dockerfile:

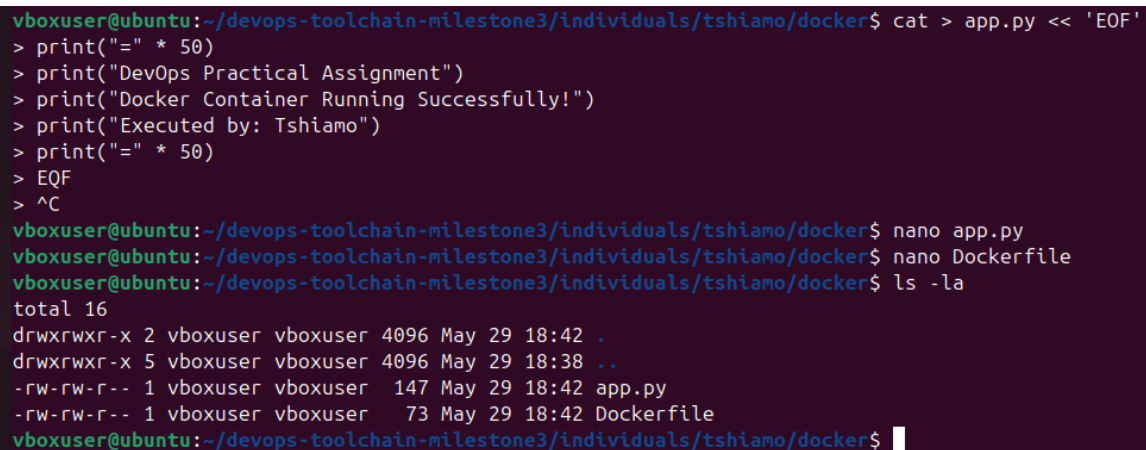
FROM python:3.9-slim

WORKDIR /app

COPY app.py .

CMD ["python", "app.py"]

Output:

A terminal window with a dark background and light green text. The user is in a directory ~/devops-toolchain-milestone3/individuals/tshiamo/docker. They use 'cat' to create app.py and 'nano' to create Dockerfile. A 'ls -la' command shows the files app.py and Dockerfile with their permissions, owners, sizes, and timestamps.

```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ cat > app.py << 'EOF'
> print("=" * 50)
> print("DevOps Practical Assignment")
> print("Docker Container Running Successfully!")
> print("Executed by: Tshiamo")
> print("=" * 50)
> EOF
> ^C
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ nano app.py
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ nano Dockerfile
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ ls -la
total 16
drwxrwxr-x 2 vboxuser vboxuser 4096 May 29 18:42 .
drwxrwxr-x 5 vboxuser vboxuser 4096 May 29 18:38 ..
-rw-rw-r-- 1 vboxuser vboxuser 147 May 29 18:42 app.py
-rw-rw-r-- 1 vboxuser vboxuser 73 May 29 18:42 Dockerfile
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$
```

Figure 7: app.py and Dockerfile created successfully

Build Docker Image

Objective: Build a custom Docker image

Command:

```
docker build -t tshiamo-python-app .
```

Output:

```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ docker build
-t tshiamo-python-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 3.072kB
Step 1/4 : FROM python:3.9-slim
3.9-slim: Pulling from library/python
ea56f685404a: Pulling fs layer
fc7443084902: Pulling fs layer
b3ec39b36ae8: Pulling fs layer
38513bd72563: Pulling fs layer
ea56f685404a: Download complete
b3ec39b36ae8: Download complete
a5e5b1b19090: Download complete
fc7443084902: Download complete
967a6079ce08: Download complete
38513bd72563: Download complete
38513bd72563: Pull complete
b3ec39b36ae8: Pull complete
fc7443084902: Pull complete
ea56f685404a: Pull complete
Digest: sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
Status: Downloaded newer image for python:3.9-slim
----> 2d97f6910b16
Step 2/4 : WORKDIR /app
----> Running in 0fa9abc855da
----> Removed intermediate container 0fa9abc855da
----> 01bcb9553cff
Step 3/4 : COPY app.py .
----> 72d9cfa31d66
Step 4/4 : CMD ["python", "app.py"]
----> Running in 46f7b37fe143
----> Removed intermediate container 46f7b37fe143
----> ee4d52f9098a
Successfully built ee4d52f9098a
Successfully tagged tshiamo-python-app:latest
```

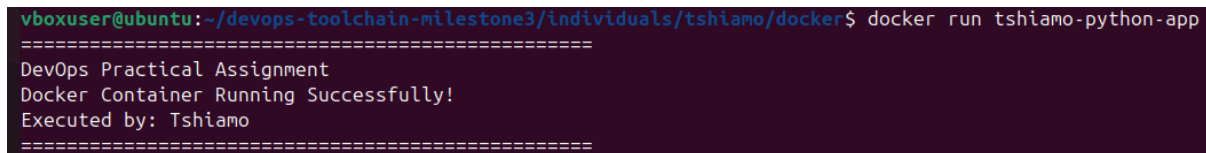
Figure 8: Docker image built successfully (python:3.9-slim pulled, layers cached)

Run Docker Container

Objective: Execute container and verify output

Command:

```
docker run tshiamo-python-app
```

A terminal window with a dark purple background. The prompt is 'vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker\$'. The command 'docker run tshiamo-python-app' has been executed. The output consists of several lines of text: '=====', 'DevOps Practical Assignment', 'Docker Container Running Successfully!', 'Executed by: Tshiamo', and '====='.

```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/docker$ docker run tshiamo-python-app
=====
DevOps Practical Assignment
Docker Container Running Successfully!
Executed by: Tshiamo
=====
```

Figure 9: Container running successfully with expected output

Commit Docker Work

Objective: Save Docker exercise to GitHub

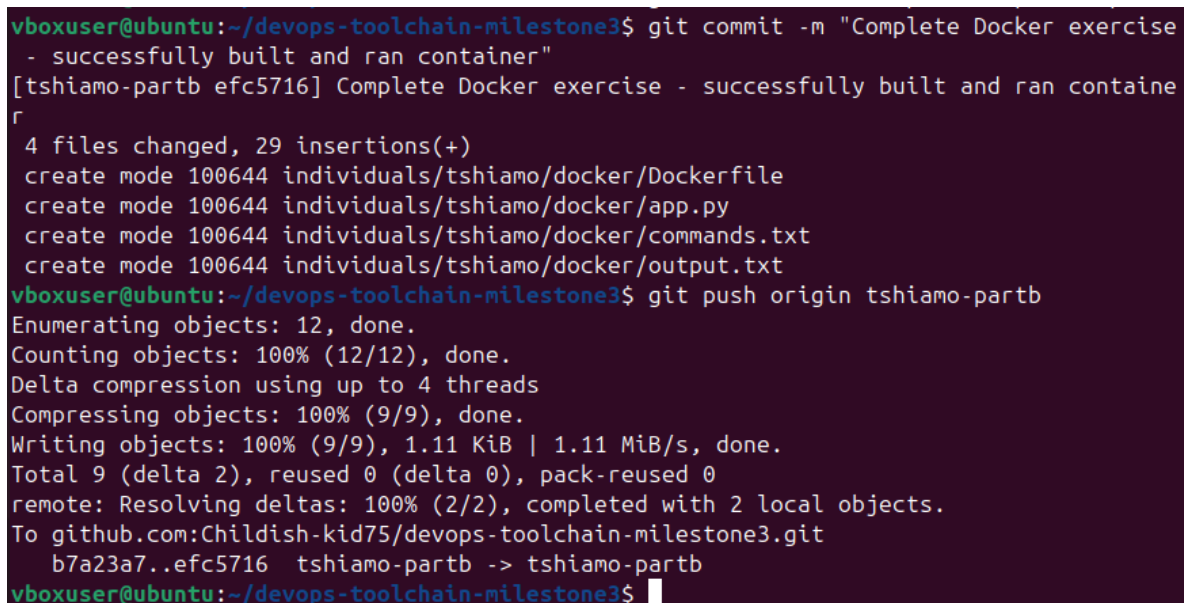
Command:

```
git add individuals/tshiamo/docker/
```

```
git commit -m "Complete Docker exercise - successfully built and ran container"
```

```
git push origin tshiamo-partb
```

Output:

A terminal window with a dark purple background. The prompt is 'vboxuser@ubuntu:~/devops-toolchain-milestone3\$'. The command 'git commit -m "Complete Docker exercise - successfully built and ran container"' has been executed. The output shows the commit details: '[tshiamo-partb efc5716] Complete Docker exercise - successfully built and ran container'. Then, the command 'git push origin tshiamo-partb' has been executed. The output shows the push details: 'Enumerating objects: 12, done.', 'Counting objects: 100% (12/12), done.', 'Delta compression using up to 4 threads', 'Compressing objects: 100% (9/9), done.', 'Writing objects: 100% (9/9), 1.11 KiB | 1.11 MiB/s, done.', 'Total 9 (delta 2), reused 0 (delta 0), pack-reused 0', 'remote: Resolving deltas: 100% (2/2), completed with 2 local objects.', 'To github.com:Childish-kid75/devops-toolchain-milestone3.git', 'b7a23a7..efc5716 tshiamo-partb -> tshiamo-partb'. The prompt is now 'vboxuser@ubuntu:~/devops-toolchain-milestone3\$'.

```
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git commit -m "Complete Docker exercise
- successfully built and ran container"
[tshiamo-partb efc5716] Complete Docker exercise - successfully built and ran containe
r
4 files changed, 29 insertions(+)
create mode 100644 individuals/tshiamo/docker/Dockerfile
create mode 100644 individuals/tshiamo/docker/app.py
create mode 100644 individuals/tshiamo/docker/commands.txt
create mode 100644 individuals/tshiamo/docker/output.txt
vboxuser@ubuntu:~/devops-toolchain-milestone3$ git push origin tshiamo-partb
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 4 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.11 KiB | 1.11 MiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:Childish-kid75/devops-toolchain-milestone3.git
b7a23a7..efc5716 tshiamo-partb -> tshiamo-partb
vboxuser@ubuntu:~/devops-toolchain-milestone3$
```

Figure 10: Docker exercise committed and pushed to tshiamo-partb branch

Git Exercise

Create Git Documentation

Objective: Demonstrate Git commands and workflow

Commands Documented:

git init

git add index.html

git commit -m "Initial commit"

git log --oneline

git status

Output:

```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ nano index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git add index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git commit -m "Initial commit: Add index.html"
[master (root-commit) 4d4952d] Initial commit: Add index.html
 1 file changed, 11 insertions(+)
 create mode 100644 index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ echo "<!-- Updated: May 2026 -->" >> index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git add index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git commit -m "Update: Add timestamp comment"
[master 5a79984] Update: Add timestamp comment
 1 file changed, 1 insertion(+)
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git log --oneline
fatal: unrecognized argument: --oneline
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git log --oneline
5a79984 (HEAD -> master) Update: Add timestamp comment
4d4952d Initial commit: Add index.html
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$
```

Figure 11: Git exercise documentation committed

Verify Git History

Objective: Show commit history

Command:

git log --oneline --graph --all

Output:

```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/git-exercise-demo$ git log --oneline --graph --all
* 5a79984 (HEAD -> master) Update: Add timestamp comment
* 4d4952d Initial commit: Add index.html
```

Figure 12: Git commit history showing all exercise commits

Chef Exercise

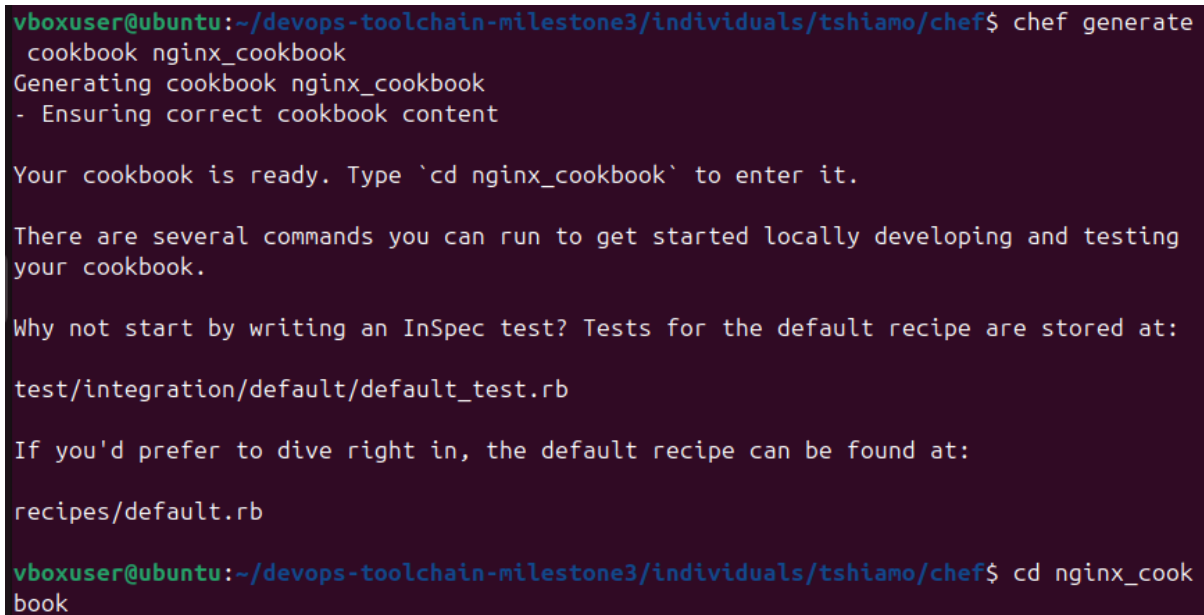
Generate Cookbook

Objective: Create Chef cookbook structure

Command:

```
chef generate cookbook nginx_cookbook
```

Output:



```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$ chef generate
cookbook nginx_cookbook
Generating cookbook nginx_cookbook
- Ensuring correct cookbook content

Your cookbook is ready. Type `cd nginx_cookbook` to enter it.

There are several commands you can run to get started locally developing and testing
your cookbook.

Why not start by writing an InSpec test? Tests for the default recipe are stored at:
test/integration/default/default_test.rb

If you'd prefer to dive right in, the default recipe can be found at:
recipes/default.rb

vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$ cd nginx_cook
book
```

Figure 13: Chef cookbook generated successfully

Create Chef Recipe

Recipe Content (nginx-recipe.rb):

```
cd ~/devops-toolchain-milestone3/individuals/tshiamo/chef
```

```
# Create a standalone recipe file
```

```
cat > nginx-recipe.rb << 'EOF'
```

```
# Install Nginx
```

```
package 'nginx' do
```

```
  action :install
```

```
end
```

```
# Start and enable Nginx service
```

```
service 'nginx' do
```

```
    action [:enable, :start]
```

```
end
```

```
# Create custom index.html
```

```
file '/var/www/html/index.html' do
```

```
  content '
```

```
    <!DOCTYPE html>
```

```
    <html>
```

```
    <head>
```

```
      <title>Tshiamo - Chef Exercise</title>
```

```
    </head>
```

```
    <body>
```

```
      <h1>DevOps Practical Assignment</h1>
```

```
      <h2>Chef Configured Nginx</h2>
```

```
      <p>Successfully configured by: Tshiamo</p>
```

```
      <p>Date: May 2026</p>
```

```
      <hr>
```

```
      <p>This web server was installed using Chef.</p>
```

```
    </body>
```

```
  </html>
```

```
  '
```

```
  mode '0644'
```

```
  action :create
```

```
end
```

```
EOF
```

```
# Apply the recipe directly
```

```
sudo chef-apply nginx-recipe.rb
```

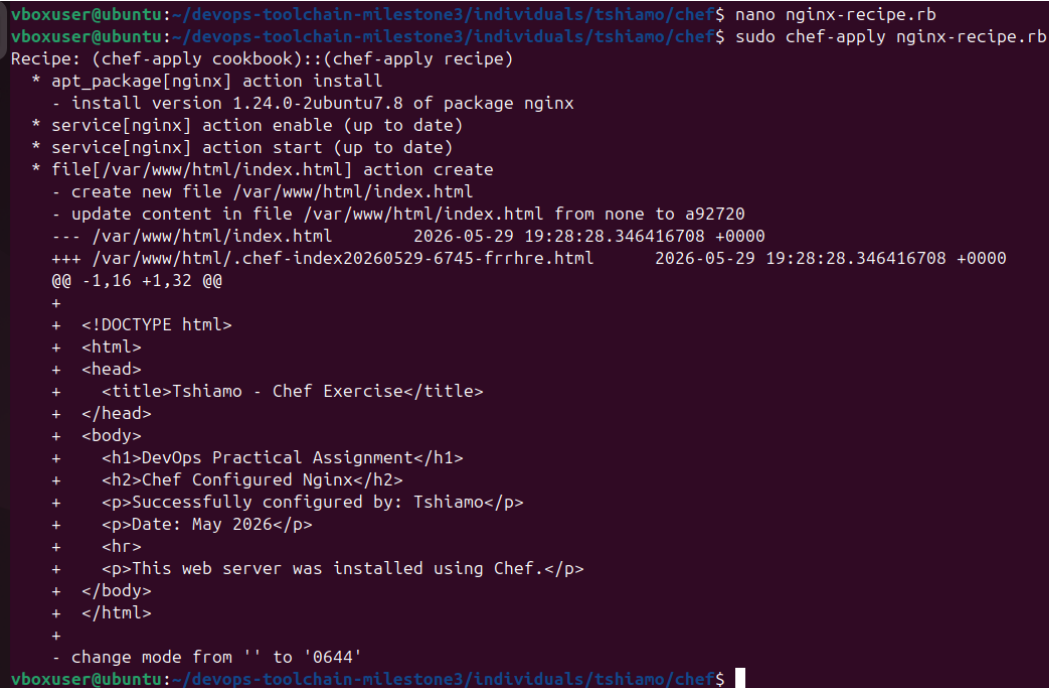
Apply Chef Recipe

Objective: Run chef-apply to configure Nginx

Command:

```
sudo chef-apply nginx-recipe.rb
```

Output:



```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$ nano nginx-recipe.rb
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$ sudo chef-apply nginx-recipe.rb
Recipe: (chef-apply cookbook)::(chef-apply recipe)
 * apt_package[nginx] action install
   - install version 1.24.0-2ubuntu7.8 of package nginx
 * service[nginx] action enable (up to date)
 * service[nginx] action start (up to date)
 * file[/var/www/html/index.html] action create
   - create new file /var/www/html/index.html
   - update content in file /var/www/html/index.html from none to a92720
   -- /var/www/html/index.html          2026-05-29 19:28:28.346416708 +0000
   +++ /var/www/html/.chef-index20260529-6745-frrhre.html      2026-05-29 19:28:28.346416708 +0000
   @@ -1,16 +1,32 @@
   +
   + <!DOCTYPE html>
   + <html>
   + <head>
   +   <title>Tshiamo - Chef Exercise</title>
   + </head>
   + <body>
   +   <h1>DevOps Practical Assignment</h1>
   +   <h2>Chef Configured Nginx</h2>
   +   <p>Successfully configured by: Tshiamo</p>
   +   <p>Date: May 2026</p>
   +   <hr>
   +   <p>This web server was installed using Chef.</p>
   + </body>
   + </html>
   +
   - change mode from '' to '0644'
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$
```

Figure 14: Chef-apply successfully installed Nginx and created custom webpage

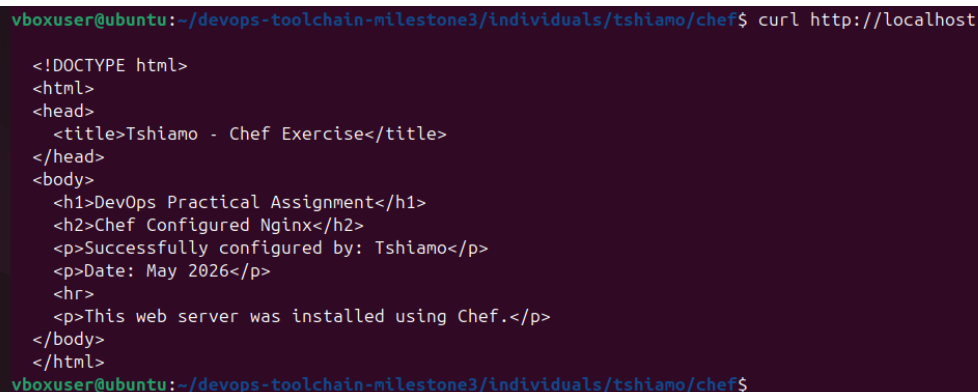
Verify Nginx Webpage

Objective: Confirm webpage is accessible

Command:

```
curl http://localhost
```

Output:



```
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Tshiamo - Chef Exercise</title>
</head>
<body>
  <h1>DevOps Practical Assignment</h1>
  <h2>Chef Configured Nginx</h2>
  <p>Successfully configured by: Tshiamo</p>
  <p>Date: May 2026</p>
  <hr>
  <p>This web server was installed using Chef.</p>
</body>
</html>
vboxuser@ubuntu:~/devops-toolchain-milestone3/individuals/tshiamo/chef$
```

Figure 15: Custom webpage successfully served by Nginx

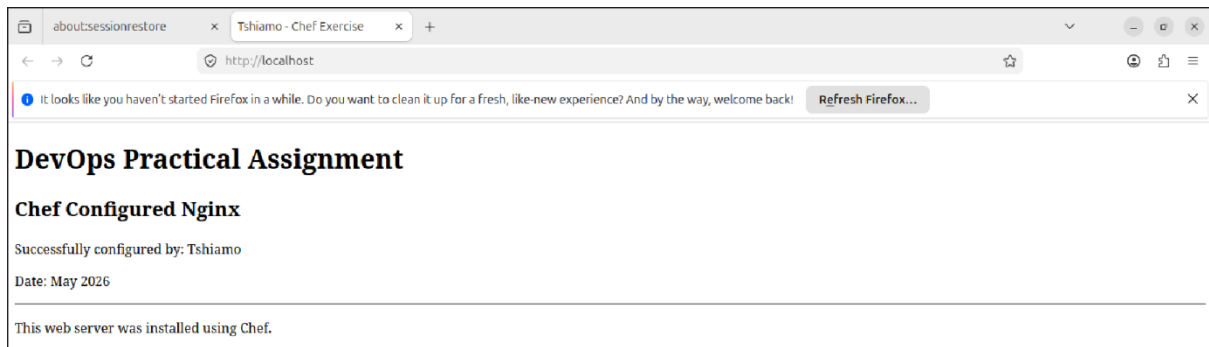


Figure 15: Custom webpage successfully served by Nginx

Final GitHub Status

All Commits Pushed

Objective: Verify all Part B exercises are on GitHub

Commits on tshiamo-partb branch:

<u>Complete Docker exercise - successfully built and ran container</u> Childish-kid75 committed 43 minutes ago	efc5716	<>
<u>Complete Git exercise documentation - Tshiamo</u> Childish-kid75 committed 35 minutes ago	adeb5c6	<>
<u>Complete Chef exercise - Nginx successfully installed and configured</u> Childish-kid75 committed 7 minutes ago	df44eeb	<>
<u>COMPLETE: All Part B exercises (Docker, Git, Chef) - Tshiamo</u> Childish-kid75 committed 3 minutes ago	2e1f436	<>
<u>Add Part B summary document</u> Childish-kid75 committed 1 minute ago	a96f1d5	<>

Figure 16: Complete commit history showing all three exercises

Commits:

1. efc5716 - Complete Docker exercise
2. adeb5c6 - Complete Git exercise documentation
3. df44eeb - Complete Chef exercise
4. 2e1f436 - COMPLETE: All Part B exercises
5. a96f1d5 - Add Part B summary document

GitHub Branch Overview

Objective: Show branch structure on GitHub

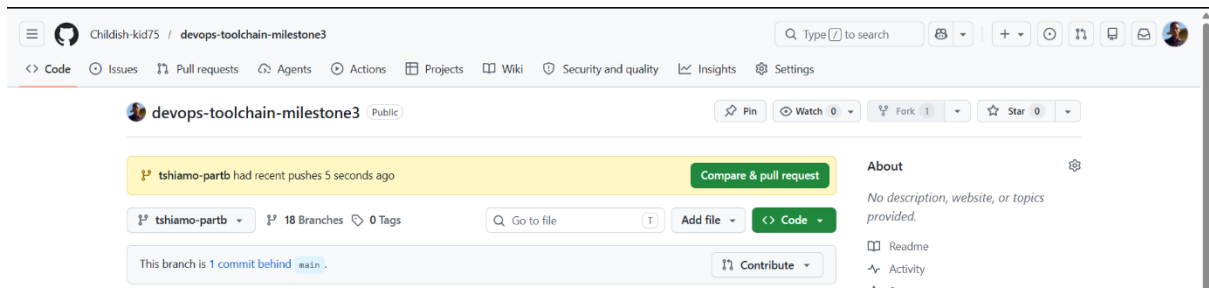


Figure 17: GitHub repository showing tshiamo-partb branch and recent pushes